

HONGHAO LIU

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Education

Hong Kong University of Science and Technology (Guangzhou) <i>Ph.D. in Artificial Intelligence, GPA: 4.2/4.3, Supervisor: Lionel Ni</i>	Sep. 2024 – present <i>Guangzhou, Guangdong</i>
Hong Kong University of Science and Technology <i>MPhil. in Data Science and Analytics, GPA: 3.7/4.3, Supervisors: Qiong Luo & Yang Xiang</i>	Sep. 2020 – Sep. 2022 <i>Hong Kong</i>
Wuhan University of Technology <i>Bachelor of Engineering in Computer Science GPA: 88.6/100</i>	Sep. 2016 – Jun. 2020 <i>Wuhan, Hubei</i>

Work Experience

International Digital Economy Academy, Research Intern Privacy-Preserving Data Synthesis for Continual LLM Pretraining - FlowHarm: a benchmark for measuring safety of agentic workflow generation and Everything will follow: generalize the workflow generation framework by enabling autonomous, self-refining to adapt to diverse, real-world tasks. - Proposed an entity-based method that synthesizes high-quality encrypted data to preserve personally identifiable information (PII) while enabling LLMs to securely encode new knowledge.	Sep. 2024 – present <i>Shenzhen, Guangdong</i>
DJI, Research Intern Accelerating perception algorithm for autonomous vehicles Implemented utility functions for the perception algorithms on the GPUs using CUDA C; Coordinated with the high-performance computing and perception teams; Applied this work to the listed Kiwi vehicles.	Jul. 2022 – Sep. 2022 <i>Shenzhen, Guangdong</i>

Research Experience

Multi-talker Speech Decoding from the Neural Activity <i>Research Assistant, The City University of Hong Kong</i> Collecting the EEG and fMRI data; Preprocessed fMRI data with the fMRIPrep and QSIPrep; Designed the multi-speaker decoding workflow, achieved BLEU-1 scores of 21% under both single- and multi- talker conditions.	Jan. 2023 - May 2024 <i>Hong Kong</i>
Efficient Radio Interferometric Imaging on the GPU <i>The Hong Kong University of Science and Technology</i> Designed and implemented and designed the non-uniform Fourier transform (NUFFT) in CUDA, parallelized the gridding and degriding methods; Optimized the memory access pattern and throughput and loading balance;	Oct. 2020 – Sep. 2022 <i>Hong Kong</i>

Publications

- [1] **H. Liu**, C. Xu, X. Jiang, C. Yang, S. Yin, Z. Ma, L. Ni and J. Guo, *Conflicts Make Large Reasoning Models Vulnerable to Attacks*, accepted by **ACL findings**, 2026.
- [2] C. Yang, X. Lin, X. Wu, C. Xu, X. Jiang, **H. Liu**, H., J. Guo *Select2reason: Efficient instruction-tuning data selection for long-cot reasoning*, accepted by **ACL findings**, 2026.
- [3] **H. Liu**, X. Jiang, C. Xu, C. Yang, Y. Cheng, L. Ni and J. Guo, *Continual Pretraining on Encrypted Synthetic Data for Privacy-Preserving LLMs*, In Findings of the ACL: **EACL 2026**, pages 408–425, Rabat, Morocco.
- [4] J. Gu, X. Jiang, Z. Shi, H. Tan, X. Zhai, C. Xu, W. Li, Y. Shen, S. Ma, **H. Liu**, S. Wang, K. Zhang, Y. Wang, W. Gao, L. Ni and J. Guo, *A Survey on LLM-as-a-Judge*, **The Innovation**, 2024, doi: 10.1016/j.xinn.2025.101253.
- [5] **H. Liu**, K. Li, Q. Zhou, Q. Wang, J. Sun, Z. Huang and J. Li, *Decoding multi-talker speech from EEG signals*, **IJCAI Workshop on Human Brain and Artificial Intelligence** (abstract), Jeju, South Korea, 2024.
- [6] **H. Liu**, Q. Luo and F. Wang, *Efficient Radio Interferometric Imaging on the GPU*, **Proceedings of IEEE 18th International Conference on e-Science**, Salt Lake City, UT, USA, 2022, doi: 10.1109/eScience55777.2022.00024.

Additional Information

Skills: python, C++, CUDA C, MPI, Linux, Git, Multi-threading, L^AT_EX, solidity
Language: Mandarin (native), English (fluent)

Awards & Prizes

- Hong Kong University of Science and Technology Scholarship
- Wuhan University of Technology Scholarship
- Wuhan University of Technology Merit Student Award

Coursework

- Parallel Programming (Teaching assistant)
- Advanced Algorithmic Techniques
- Computational Neuroscience (Summer school)
- Large Language Models: Theory and Practice